

Orientation and report writing

Fall 2026 · Prof. Ehsan Roohi
September 9, 2026

**MEASURE
MODEL
COMPARE**

Experimental evidence and engineering judgment

01

OBSERVE

See the physical behavior
worth explaining

02

MEASURE

Acquire data with a
deliberate measurement
system

03

COMPARE

Test data against theory
and models

04

COMMUNICATE

Write a report another
engineer can trust

The goal is not merely to collect numbers. The goal is to make a claim that the data support.

Today's 50-minute session

- 01** 0–8 min: course organization and workflow

- 02** 8–23 min: experiments, apparatus, and analysis

- 03** 23–42 min: report writing and comparison

- 04** 42–48 min: policies and report exercise

- 05** 48–50 min: first dates and preparation

People, places, and weekly rhythm

COURSE TEAM

Prof. Ehsan Roohi
Yuyuan Zhang
Nazila Emamdoost

WEEKLY RHYTHM

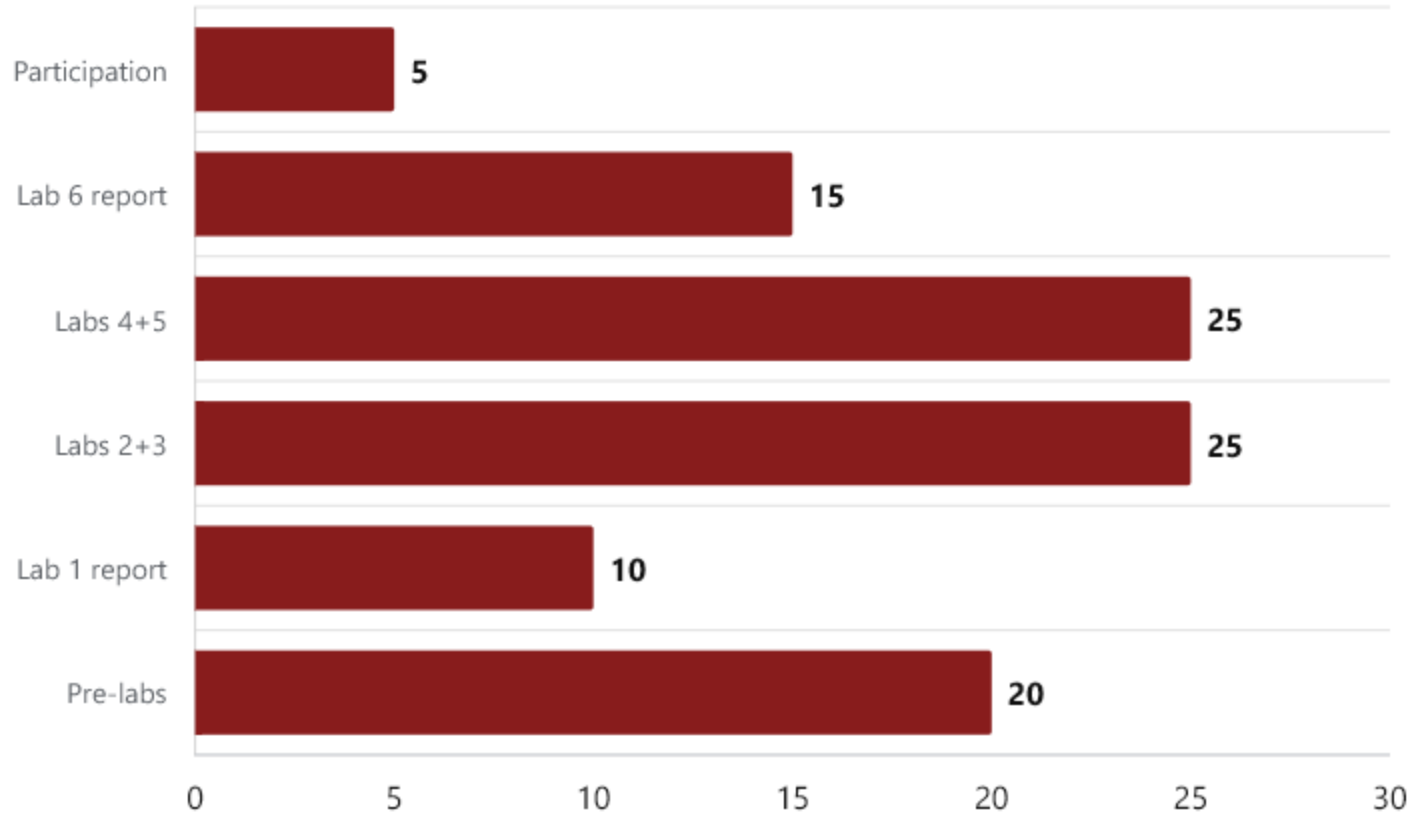
MON Theory • 12:20–1:10 PM
Hasbrouck Laboratory 134

TUE–FRI Laboratory sections
Gunness Laboratory 5

FRI Office hours • 12:20–1:10 PM
Gunness Laboratory 1

Check Canvas for your assigned section, GTA, and any section-specific updates.

Course grading

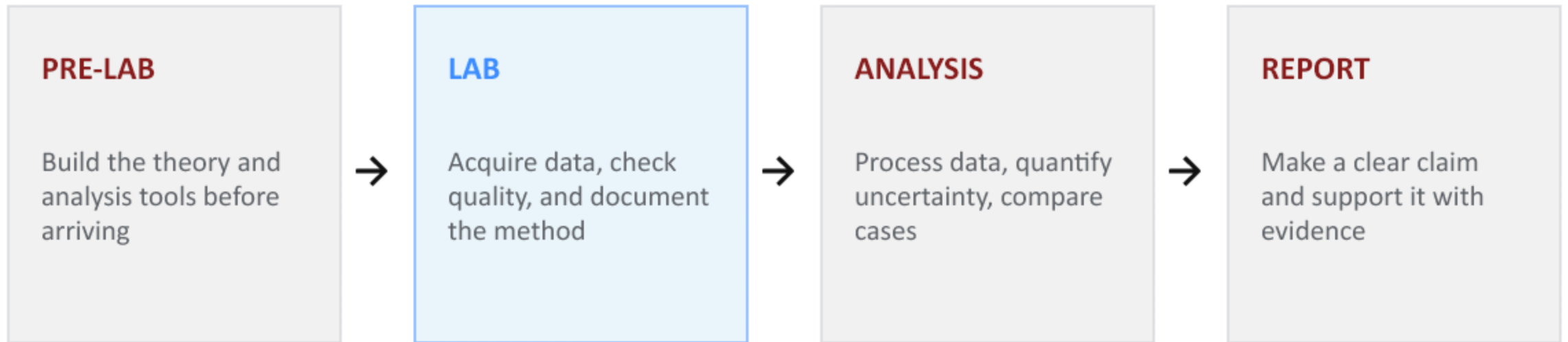


100%

**Preparation + reports +
participation**

The combined reports reward
synthesis across related
experiments.

The laboratory workflow



Pre-labs are due before your own laboratory section begins.

Six experiments and four reports

Lab	Experiment	Primary focus	Report
1	Dynamic data sampling	Sampling, aliasing, spectra	Lab 1
2	Single pendulum	Dynamic response, model–data	Labs 2+3
3	Double pendulum	Coupled and nonlinear motion	Labs 2+3
4	1-DOF free response	Natural frequency, damping	Labs 4+5
5	1-DOF forced response	Resonance, frequency response	Labs 4+5
6	Radiometer	Movement measurement	Lab 6

Lab 1: Recording sound

The question

How often must we measure a sound to record its frequency correctly?

The equipment

The microphone turns sound into voltage.
Moku:Go records that voltage as numbers.

What you do

Record several sounds. For a steady tone, repeat the recording at different sampling rates.



Moku:Go data logger connected to a microphone

Lab 1: A recording can show the wrong frequency

Two useful plots

A waveform shows voltage versus time. A frequency spectrum shows which frequencies are present; MATLAB can calculate it with an FFT.

What to look for

With too few samples per cycle, the signal can appear to have a lower frequency. This is called aliasing.

What goes in the report

Compare recordings with their actual sampling rates and durations. Explain which settings gave a reliable frequency estimate.

Saving as MAT makes the measurements easy to load in MATLAB. Changing the file format does not fix a poor recording.

Lab 2: One pendulum

The question

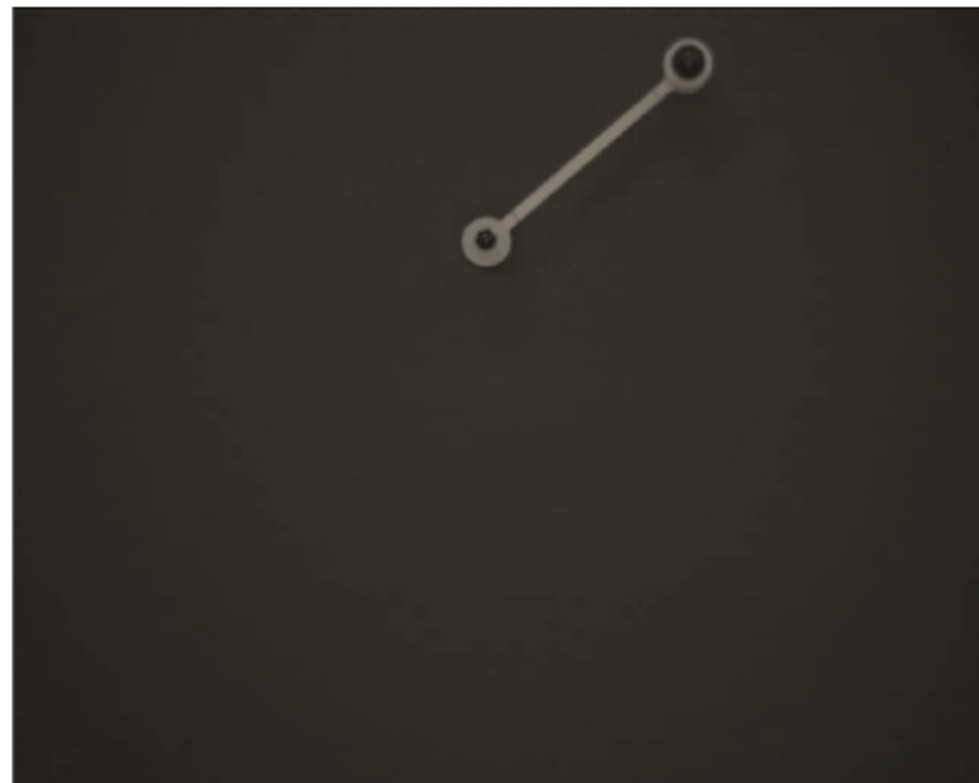
Does a pendulum take the same time to swing when we release it from a larger angle?

The equipment

A pendulum, a high-speed camera, lighting, and a length reference. The camera records the moving rod.

What you do

Release it without pushing. Record small, medium, and large starting angles. Track its position in MATLAB.



Single pendulum recorded by the high-speed camera

Lab 2: Comparing the swing with a model

The measurements

Plot angle versus time. Measure the period over several swings and check whether the swing size decreases.

The model

The small-angle model simplifies the equation of motion. Compare it with the full model as the release angle increases.

What goes in the report

Overlay measured and predicted angles. Compare their periods and explain how the starting angle changes the agreement.

Use the same starting angle, angle direction, and time origin in the measurement and simulation.

Lab 3: Two connected pendulums

The question

How does adding a second moving link change the motion and our ability to predict it?

The equipment

Two connected links, a high-speed camera, and a tracking app that follows both moving joints.

What you do

Record one small-angle case and at least three large-angle cases. Write down both starting angles before each release.



Two-link pendulum during release

Lab 3: Two angle records to explain

The measurements

Plot each link's angle versus time. Notice how the two links speed up, slow down, and influence one another.

The comparison

Start the simulation with the measured angles. Compare both links over the same time interval.

What goes in the report

Show where the predictions agree and where they separate from the measurements. Check release conditions and tracking before explaining the difference.

A small difference at release can grow during large motion. An irregular-looking trajectory alone does not prove chaos.

Lab 4: A cart released between springs

The question

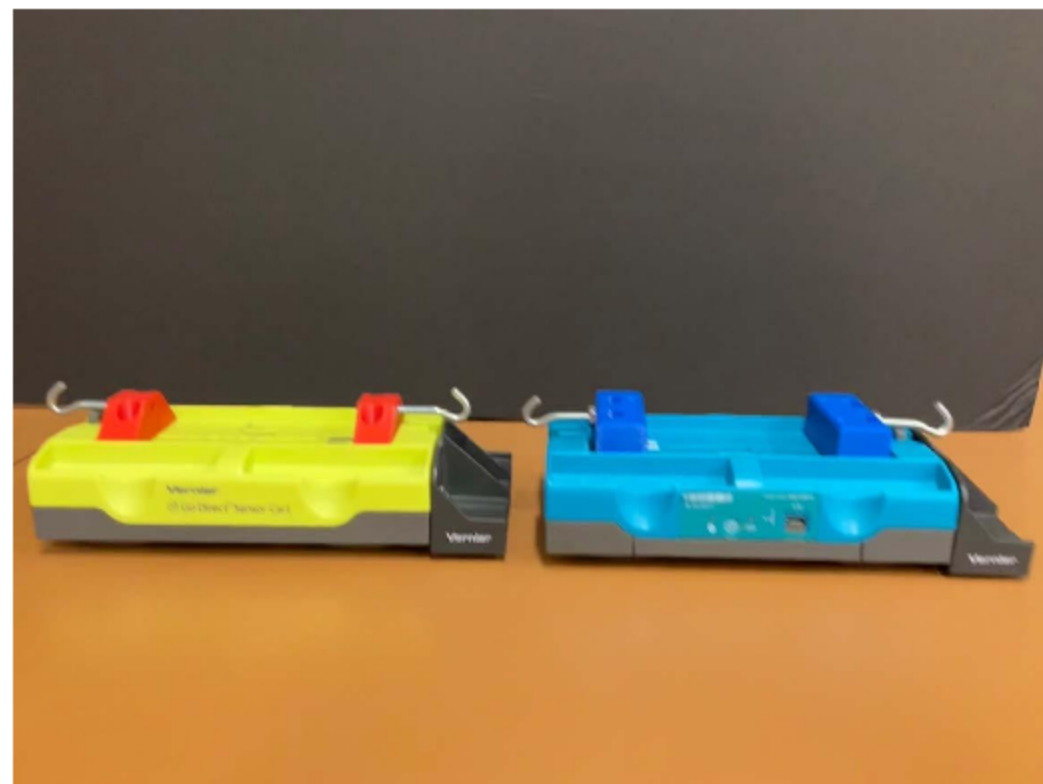
What controls how fast the cart moves back and forth, and how quickly that motion dies away?

The equipment

A cart on a track, springs, added masses, and magnetic brakes. A sensor records the cart position.

What you do

Measure mass and spring stiffness. Pull the cart away from rest and let go. Repeat after changing one setting at a time.



Vernier carts with added masses

Lab 4: Swing time and fading motion

Two features of the plot

Peak spacing gives the period. The decrease in peak height shows how quickly the motion dies away.

Expected trends

More mass usually gives slower motion. Stiffer springs give faster motion. More damping makes the peaks decrease faster.

What goes in the report

Compare the cases using measured frequency and damping. State exactly what changed and connect each trend to the model.

Keep other settings fixed when making a comparison. Include the mass of added brake parts in the total moving mass.

Lab 5: A cart driven by a motor

The question

Why can the cart move much farther at some driving speeds than at others?

The equipment

A motor moves one spring end back and forth. The cart records position; a tachometer measures motor speed.

What you do

Compare three brief pushes. Then drive the cart at several speeds below and above its strongest response.



Motor and spring drive from an earlier course setup

Lab 5: Response at different driving speeds

After each speed change

Wait until the motion settles into a repeating pattern. Measure the spring-end motion and the cart motion.

The main graph

Plot cart amplitude divided by input amplitude against driving frequency. A peak indicates resonance.

What goes in the report

Compare the measured curve with the model. Explain how mass, spring stiffness, and damping affect the response.

Amplitude means the distance from the middle position to a peak. Keep peak and peak-to-peak measurements consistent.

Lab 6: A radiometer under light

The question

How fast do the vanes turn under the light, and does that speed change as the device warms?

The equipment

A glass radiometer, LED light, light meter, fixed camera, and dark background.

What you do

Keep the setup still. Record a clear 60–90 second video and note light level, distance, and whether the device is already warm.



Radiometer and light meter; green rings mark the tracking region

Lab 6: Rotation speed from video

The measurement

The program follows the vanes and adds up their rotation angle. The slope of angle versus time gives average speed.

The useful comparison

Compare the early and late parts of the same recording. Check whether the rotation speeds up, slows down, or stays nearly steady.

What goes in the report

Report speed in revolutions per minute (RPM), show the angle plot, and explain how lighting, warming, and tracking affect the result.

Compare the same time window in each trial. Lux measures visible light level; it does not directly measure the heat absorbed.

The structure of a laboratory report

REQUIRED SECTIONS

Abstract

1. Introduction

2. Method

3. Results

4. Comparison

5. Discussion

6. Conclusions

References • Appendices

WHAT THE READER SHOULD FIND

- A clear purpose and relevant theory
- A reproducible method
- Readable figures with units and labels
- Direct comparison with theory or a model
- A discussion of discrepancies and limitations
- Conclusions supported by the presented data

Meaningful results matter more than a polished page with weak analysis.

Report length and section purpose

Section	Template allocation	Main purpose
Abstract	Maximum 10 lines	Purpose, method, key finding, implication
Introduction	1 page	Theory, experiment, and testable objective
Method	1 page	Enough detail to reproduce the measurement
Results	3 pages including figures	Observed behavior and processed measurements
Comparison	1 page	Prediction and experiment on the same basis
Discussion	1 page	Discrepancies, uncertainty, and limitations
Conclusions	1 page	Quantitative answer and the main lesson

References and optional appendices have no stated page limit. Follow any assignment-specific limit on Canvas.

Introduction and method

Introduction

Explain what you want to test, why it matters, and which equation predicts the result. Define every symbol.

Method

Describe the equipment, calibration, recording settings, and what you changed between trials.

Data processing

Explain how you turned the raw measurements into your final numbers and plots. Another group should be able to repeat the steps.

Example objective: test how release angle changes agreement with the small-angle pendulum model.

An abstract with a clear numerical result

Purpose and method

We tested a small-angle pendulum prediction using angle histories from video.

Key result

The measured frequency was 1.47 Hz, compared with a predicted 1.50 Hz, a 2.0% difference.

Meaning and limit

The agreement supports the model for this case. The discrepancy requires an uncertainty assessment.

Illustrative wording and invented teaching values. An actual abstract must use your own results. Maximum 10 lines.

Results: what the measurements show

Observation

Describe the measured trend. Give the important values and their units.

Evidence

Use a figure or table that answers the question. State the test conditions and identify repeated trials.

Origin of each curve

Label measurements and simulations clearly. Explain any processing that changes the measured values.

Example observation: successive positive displacement peaks decrease over time.

Readable, exported figures

DO

- Export high-resolution PNG, EPS, or PDF
- Label axes and include units
- Use legends that identify each data set
- Group related cases when comparison improves the analysis

AVOID

- Screenshots of MATLAB or Python figure windows
- Tiny labels and unreadable legends
- Decorative plots that do not answer a question
- Separate figures when one comparison plot is clearer

A figure should still make sense when viewed without the surrounding paragraph.

A figure that supports a comparison

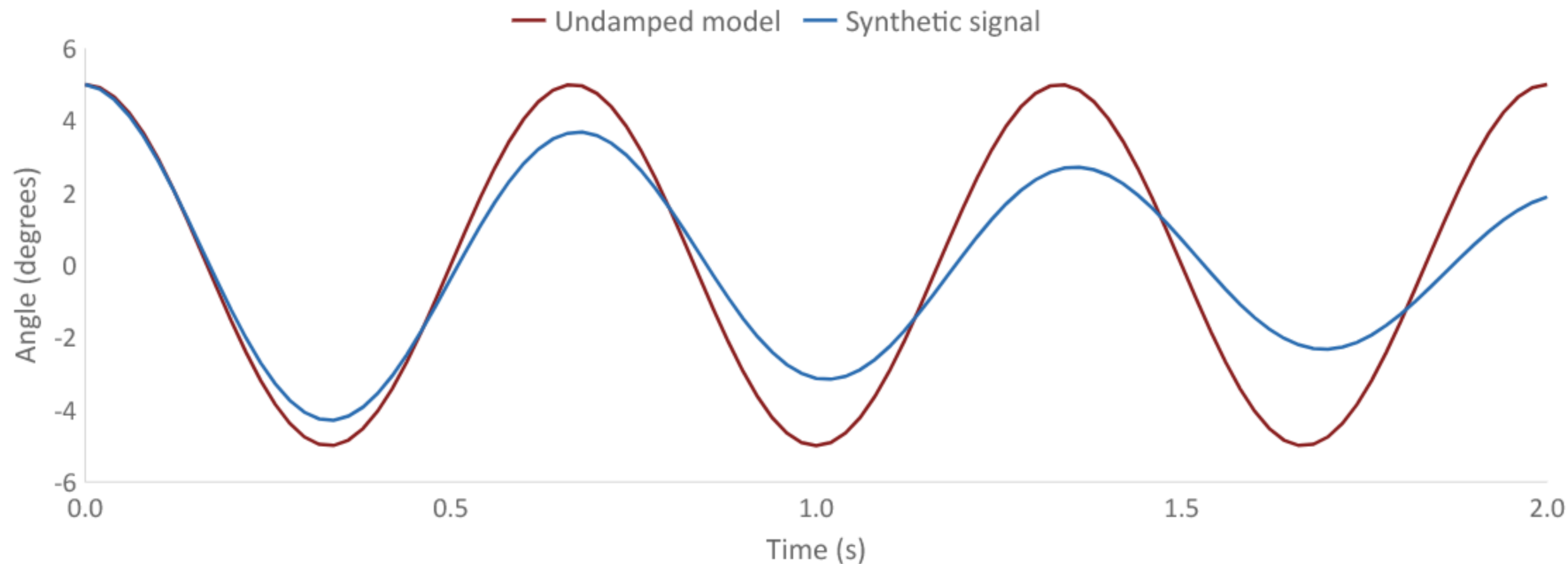


Figure 1. Illustrative angle histories: undamped model and synthetic decaying signal, 5° initial angle, 50 samples/s.

Comparison: a fair test of the model

Same conditions

Use the same units, starting conditions, and time origin. State the model parameters.

Useful numbers

Compare a relevant quantity, such as period, frequency, or amplitude. Also show the curves together when useful.

Where parameters came from

State which values you measured, assumed, or adjusted to match the data. Test adjusted values on another trial when possible.

$$\text{Relative discrepancy} = | \text{measured} - \text{predicted} | / | \text{predicted} | \times 100\%$$

Use an appropriate absolute or scaled metric when the predicted value is zero or very small.

How close is close enough?

Limits of the measurements

A camera time setting, ruler scale, or tracking position can be slightly wrong. Repeated trials can also give different results.

Limits of the model

Mass or spring stiffness may be uncertain. A simple model may leave out friction or large-angle effects.

A supported conclusion

A 2% difference alone does not tell us whether the agreement is good. Compare it with the size of the measurement and model uncertainties.

Period = elapsed time / number of complete cycles

Timing several cycles reduces the effect of choosing one peak slightly early or late.

Discussion: explanations tied to evidence

What you observed

The measured swing gets smaller over time, while the model keeps the same swing size.

A possible explanation

Friction or air resistance could remove energy. This plot alone does not tell us how much each one contributes.

A useful check

Compare how quickly the motion fades in different setups. Test whether adding damping improves the prediction.

Replace “human error” with a specific mechanism, its expected effect, and a way to check it.

Conclusions, references, and appendices

Conclusions

Answer the objective with the key numerical result, supported limitation, and a useful improvement.

References

Cite sources where used. Include the lecture notes at minimum and list every cited source.

Appendices

Add supporting calculations or extra data only when needed. Keep essential evidence in the main text.

Combined reports connect related experiments

Labs 2 and 3

Compare one angle with two coupled angles. Discuss model assumptions, initial conditions, and the limits of prediction.

Labs 4 and 5

Relate free-response frequency and damping to the forced response. Account for configuration changes.

Organization

Use one abstract and conclusion, with experiment-specific subsections and an explicit synthesis.

Suggested organization within the required template. Do not assume the page allowance doubles.

The report rubric: ten equal criteria

Each criterion is 10% of the report grade	What to check
Complete sections / overall aesthetic	Required organization and consistent formatting
Clear figures / abstract	Readable evidence and a concise, quantitative summary
Data analysis / introducing the theory	Traceable processing and defined governing equations
Introducing the experiment / comparison with theory	Physical context and a fair model-data comparison
Discussion / meaningful results	Evidence-based interpretation and an answer to the objective

This table groups the ten original rubric criteria into pairs. Each individual criterion remains worth 10%.

Submission and integrity policies

DEADLINES

No late pre-labs.
Reports lose 25% for each day or partial day late.

MAKE-UPS

Require instructor/GTA availability, space, and the correct equipment setup.

INTEGRITY

Discuss approaches, but do not copy solutions or share MATLAB, Python, or Simulink files.

VERIFICATION

Any pre-lab or report may include a brief oral check of your understanding.

Notify the instructor during the first two weeks about approved conflicts or accommodation needs.

Three-minute report exercise

Draft sentence

“The frequency was 1.47. Theory was 1.50. The error was small because of human error.”

Work with a neighbor

Add units, calculate the discrepancy, and separate the observation from its interpretation.

Then decide

What information is missing before you can judge agreement? Suggest one check that would help.

Invented teaching values. No submission is required for this in-class discussion.

Exercise debrief

Results and comparison

The measured frequency was 1.47 Hz and the predicted frequency was 1.50 Hz. Their difference was 0.03 Hz, or 2.0% of the prediction.

Discussion

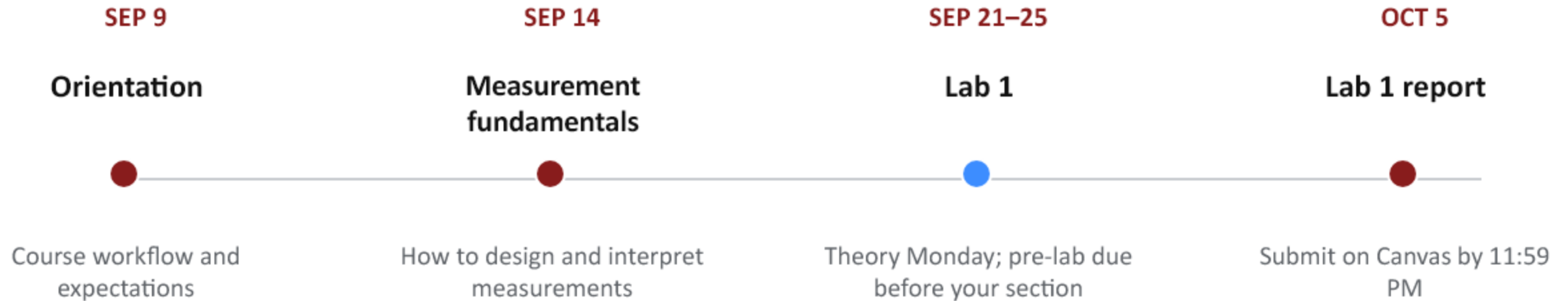
This discrepancy alone does not establish acceptable agreement. Timing and model-parameter uncertainties are needed.

Useful next check

Verify the time calibration and repeat the frequency estimate using a stated interval and method.

Other specific, evidence-based checks can also be valid.

The first dates



There is no laboratory in Weeks 1–2.

Preparation for Lab 1

BEFORE WEEK 3

- Confirm your laboratory section and GTA on Canvas
- Install or update MATLAB and Simulink
- Open the Lab 1 instructions and Pre-lab 1
- Know where and when your section meets
- Bring a way to save your experimental data

ONE QUESTION

**What would make
you trust an
experimental result?**

Questions & discussion